Task 3: Automate Application Deployment

Objective

To automate application deployment using Jenkins and a deployment script.

Deployment Script

A shell script named `deploy.sh` was created inside the `deployment` directory.

The script performs the following operations:

1. Displays the deployment start message.
2. Executes the Python application.
3. Displays the deployment completion message.

The script was made executable using:

```
chmod +x deployment/deploy.sh
```

It was executed using:

```
./deployment/deploy.sh
```

The output confirmed successful deployment.

Jenkins Deployment

The Jenkinsfile contains a dedicated Deploy stage:



The Jenkins pipeline also contains post-build actions for both successful and failed executions.

Observation

Application deployment was successfully automated through Jenkins. Once the build and test stages completed successfully, Jenkins automatically executed the deployment script.

```
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ sudo apt install python3-pytest -y
Installing:
  python3-pytest

Installing dependencies:
  python3-iniconfig  python3-pluggy

Summary:
  Upgrading: 0, Installing: 3, Removing: 0, Not Upgrading: 160
  Download size: 305 kB
  Space needed: 1545 kB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-iniconfig all 2.1.0-2 [6962 B]
Get:2 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pluggy all 1.6.0-2 [21.1 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/universe amd64 python3-pytest all 9.0.2-4 [277 kB]
Fetched 305 kB in 5s (57.4 kB/s)
Selecting previously unselected package python3-iniconfig.
(Reading database ... 38075 files and directories currently installed.)
Preparing to unpack .../python3-iniconfig_2.1.0-2_all.deb ...
Unpacking python3-iniconfig (2.1.0-2) ...
Selecting previously unselected package python3-pluggy.
Preparing to unpack .../python3-pluggy_1.6.0-2_all.deb ...
Unpacking python3-pluggy (1.6.0-2) ...
Selecting previously unselected package python3-pytest.
Preparing to unpack .../python3-pytest_9.0.2-4_all.deb ...
Unpacking python3-pytest (9.0.2-4) ...
Setting up python3-iniconfig (2.1.0-2) ...
Setting up python3-pluggy (1.6.0-2) ...
Setting up python3-pytest (9.0.2-4) ...
Processing triggers for man-db (2.13.1-1build1) ...
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 -m pytest --version
pytest 9.0.2
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ python3 -m pytest
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/admin1/devops-ci-cd-lab
plugins: typeguard-4.4.4
```

```
============================= test session starts ==============================
platform linux -- Python 3.14.4, pytest-9.0.2, pluggy-1.6.0
rootdir: /home/admin1/devops-ci-cd-lab
plugins: typeguard-4.4.4
collected 1 item

test_app.py .

============================== 1 passed in 0.01s ===============================
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ mkdir -p deployment
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ nano deployment/deploy.sh
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ chmod +x deployment/deploy.sh
admin1@DESKTOP-BBG39NK:~/devops-ci-cd-lab$ ./deployment/deploy.sh
Starting deployment...
Hello, Gurleen!
Deployment completed successfully.
```

